

Daily Scholar Papers Report — 2026-07-11

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Window covered: 2026-07-10 → 2026-07-11 (Google Scholar alerts + user-curated self-emails, last 24 h)

Executive Summary

Two Scholar-alert threads landed in the window — one for Bihuan Chen (four candidates) and one for Yangruibo Ding (a single ICML position paper) — and the user-curated forward queue was empty. Four papers cleared Stage-1; one arXiv text-to-video training-orchestration paper was routed as a name collision (a different “B Chen” than the Fudan SE researcher) and dropped without deep-read. Two of the four proceeds sit behind the ACM digital library with no arXiv mirror on hand, so they are analysed at abstract depth and marked Keep; the other two (a TOSEM/arXiv paper on autonomous-driving-scenario testing and an ICML position paper on offensive-security AI agents) had full text available and are analysed as Outstanding. Preference-layer signal is quiet — `venue::preprint` is the only attribute past its $n \geq 2$ threshold (posterior 0.75, mild tailwind for the two arXiv-hosted proceeds), and no candidate crossed the 0.30 auto-Skip threshold or needed the preference-promoted override.

Outstanding: 2 · **Keep:** 2 · **Borderline High-Priority:** 0

Highlighted Papers

#	Title	Authors	Venue	Link
1.1	DynNPC: Finding More Violations Induced by ADS in Simulation Testing through Dynamic NPC Behavior Generation	Y. Lu, Y. Tian, D. Wang, B. Chen, X. Peng	TOSEM / ICSE 2026, also arXiv 2411.19567	arXiv
1.2	Position: To Defend Against Cyber Attacks, We Must Teach AI Agents to Hack	T.Y. Zhuo, Y. Ding, W. Guo, R. Meng	ICML 2026 Position, also arXiv 2602.02595	OpenReview
2.1	Understanding Performance		PACMSE (FSE) 2026	DOI

#	Title	Authors	Venue	Link
	Problems in CUDA Programs	Y. Bi, J. Cao, Y. Lu, B. Chen, T. Gan, D. Wang, X. Peng		
2.2	It Takes Two: Option-Aware Directed Greybox Fuzzing for Vulnerability PoC Generation	S. Wu, X. Hu, Y. Cao, Z. Zhou, Y. Huang, Y. Wu, B. Chen, Z. Zhao, X. Peng	PACMSE (FSE) 2026	DOI

1. Outstanding

1.1 · ADS SCENARIO TESTING · Dynamic in-simulation NPC controllers with three driving strategies raise Ego-induced-violation share by an average 72.18% and cut time-to-first-EIV by 90.87% over five ADS-testing baselines

1.1 [DynNPC: Finding More Violations Induced by ADS in Simulation Testing through Dynamic NPC Behavior Generation](#) — Y. Lu, Y. Tian, D. Wang, B. Chen, X. Peng — ACM Transactions on Software Engineering and Methodology (TOSEM), arXiv 2411.19567v2

Venue. TOSEM (top-tier SE journal); to be presented at ICSE 2026 in Rio de Janeiro, April 12–18, 2026.

Problem. Existing scenario-based testing tools for autonomous driving systems (ADSs) — the paper positions itself against AV-Fuzzer, AutoFuzz, CRISCO, BehAVExplor and DoppelTest — predefine NPC vehicle behavior sequences before simulation execution and only mutate them across generations of the search. That approach ignores both traffic-signal state and the real-time behavior of the Ego vehicle, so many of the reported “violations” turn out to be caused by unrealistic NPC behavior rather than genuine ADS bugs. The authors reference prior work reporting 1,109 automatically generated violation scenarios collected in 240 hours whose violations were later attributable to NPC-induced (not ADS-induced) failure — a stark efficiency ceiling.

Method. A dynamic NPC controller selects a maneuver decision per simulation step conditioned on traffic-signal state and current Ego trajectory, drawing from a three-strategy menu: *yielding* (defensive), *adversarial* (attempts near-miss scenarios), and *overtaking* (aggressive lane changes). An automatic trajectory generator then materialises the chosen decision as a physically realisable path. Scenario configurations — map, spawn points, initial NPC roles, Ego task specification — are searched by a genetic-algorithm scenario generator, and a new scenario executor drives each configuration in LGSVL 2021.3 against Apollo 7.0 under a 12-hour experiment budget.

Headline numbers (verbatim from paper). Ego-induced violation (EIV) proportion increase averaged **72.18%** across the five baselines; unique EIV pattern coverage improvement **at least 38.71%**; time-to-first-EIV reduction **90.87%**; average time-per-EIV reduction **81.85%**. The paper also reports that the realism of generated NPC vehicle speed sequences is improved.

Reusability. The “three-strategy maneuver menu + genetic-algorithm scenario search + independent scenario executor” trio is a clean reference architecture for any simulation-based fuzz-testing pipeline where the environment is not a static input but a set of adversarial agents. The maneuver template is small enough that it can plausibly be ported to drone-swarm testing or multi-agent robotics scenarios by swapping the strategy menu.

Cross-paper connection. Pairs naturally with the ADS-testing work summarised in the [2026-06-24 digest](#) — DynNPC is the most direct answer yet to the “NPC-behavior realism ceiling” that thread has been flagging.

Closing citable line: “DynNPC outperforms in the realism of generated NPC vehicle speed sequences.”

1.2 · AI AGENTS & CYBER OFFENSE · Position paper argues model-centric AI safety cannot stop AI-agent-driven cyber attacks and proposes a three-tier offense-informed defensive agenda anchored on audited cyber ranges and offense-to-defense distillation

1.2 [Position: To Defend Against Cyber Attacks, We Must Teach AI Agents to Hack](#) — T.Y. Zhuo, Y. Ding, W. Guo, R. Meng — ICML 2026 (Position track), arXiv 2602.02595

Venue. ICML 2026 Position paper — a curated track that is the ML community’s closest analogue to the SP magazine position genre.

Problem. Cybersecurity’s operating equilibrium has long relied on human labor scarcity: sophisticated exploits are expensive to build, which forces attackers to concentrate on high-value targets and leaves most systems protected by obscurity and cost. AI agents disrupt that equilibrium by automating the vulnerability-discovery → exploit-development → deployment loop; they need only small per-target success rates to remain profitable when applied at scale. The authors argue that the standard model-centric AI-safety toolkit — training-data governance, safety alignment, representation engineering, output guardrails, access controls — cannot hold this line once adversaries have open-weight models or roll their own offensive capability.

Threat model. A financially motivated, technically sophisticated attacker constrained mainly by labor cost. Accesses SOTA agents via APIs or local deployment. Objective is aggregate profit across heterogeneous victims, not damage to a single high-value target. Low per-attempt success rates remain economically viable at massive scale. Nation-state capabilities and novel cryptographic breaks are explicitly out of scope.

Argument. The paper proposes an offense-informed defensive agenda in three parts. First, **frontier offensive security measurement** — build comprehensive attack-lifecycle benchmarks aligned to MITRE / cyber-kill-chain phases rather than the current patchwork of siloed benchmarks. Second, **frontier offensive security development** — move from workflow-scaffolded agents (LLM plus tool orchestration) to trained offensive agents optimised by reinforcement learning inside verifiable cyber ranges. Third, **frontier offensiveness protection** — capability-tiered checkpoints, version-controlled staged release, audited cyber-range containment, and offense-to-defense distillation, so the surface exposed to the public is a defensive-only agent trained from the offensive one rather than the raw capability itself.

Headline numbers (verbatim from paper Table 1 — SOTA agent scores across offensive-security benchmarks). CyberSecEval-3 (attack generation) **49%**, SeCodePLT (attack generation) **0.2%**, AutoPenBench (attack generation) **54.5%**, CVE-bench (attack generation) **12.5%**, CyBench (CTF) **55%**,

NYU CTF **22%**, PrimeVul (vuln-detection) **12.9%**, VulnLLM (vuln-detection) **77.8%**, CyberGym (PoC generation) **28.9%**, SEC-bench (patching) **22.3%**, SWE-bench-Verified (patching) **78.8%**. The spread — 0.2% on one attack-generation benchmark to 78.8% on a patching benchmark — is itself the paper’s central empirical claim: benchmarks disagree wildly, so a single-number “agent capability” figure is meaningless without lifecycle-phase decomposition.

Reusability. The three-tier framework (measure → develop → protect) is a useful scaffold for anyone writing agenda-level papers on dual-use agentic systems; it generalises well beyond cybersecurity to biosecurity, autonomous-weapons, and financial-fraud settings that share the same “aggregate profit across many low-value targets” adversary model. The benchmark scoreboard is a ready reference for anyone starting a new offensive-security agent project.

Closing citable line: “Offensive AI capabilities should be treated as essential defensive infrastructure.”

2. Keep

2.1 · GPU PERFORMANCE · Empirical companion to the group’s earlier DL-systems performance study, retargeted to raw CUDA programs and evaluating existing performance analysis tools against a reproduced defect set

2.1 [Understanding Performance Problems in CUDA Programs](#) — Y. Bi, J. Cao, Y. Lu, B. Chen, T. Gan, D. Wang, X. Peng — Proceedings of the ACM on Software Engineering (PACMSE) 2026

Venue. PACMSE / FSE 2026.

Positioning. Reads as the CUDA-native successor to the same group’s earlier [Understanding Performance Problems in Deep Learning Systems](#) (ISSTA 2022). Where that paper characterised performance defects at the TensorFlow / Keras layer, this one drops down to raw CUDA — where the complex programming model (memory hierarchy, warp-level primitives, launch-configuration knobs, host/device data movement) creates a fresh symptom and root-cause taxonomy that DL-framework studies do not surface.

Contribution shape (from the paper’s Scholar snippet and ACM landing metadata). A symptom-and-root-cause characterisation of CUDA performance defects mined from developer forums and issue trackers, followed by a measurement of how well existing performance-analysis methods identify the reproduced problems. The full defect count, taxonomy leaves, and tool-effectiveness numbers require the ACM PDF — direct fetch was blocked and no arXiv mirror surfaced today.

Why it stays on the radar. The methodology (mining developer forums for real performance complaints, reproducing them, and testing whether existing analyses catch them) is exactly the kind of empirical scaffolding that seeds subsequent tool-building projects. Even at abstract depth, it is worth tracking for anyone working on GPU performance analysis, tuning agents, or LLM-assisted CUDA optimisation. A re-read is queued once the paper text becomes retrievable.

2.2 · DIRECTED FUZZING · Joint option × input search inside the directed-greybox-fuzzing loop — the “it takes two” framing signals that reaching a vulnerable location needs both the right input and the right command-line option combination

[2.2 It Takes Two: Option-Aware Directed Greybox Fuzzing for Vulnerability PoC Generation](#) — S. Wu, X. Hu, Y. Cao, Z. Zhou, Y. Huang, Y. Wu, B. Chen, Z. Zhao, X. Peng — PACMSE / FSE 2026

Venue. PACMSE / FSE 2026 (Montreal).

Problem framing (from Scholar snippet). Static analysis produces bug candidates but rarely proofs-of-concept. Directed greybox fuzzers (DGF) narrow the search to a target location, but they typically treat command-line options as fixed. The paper’s “it takes two” framing signals that reaching a vulnerable location in many real targets requires both the right input and a specific option combination — an insight adjacent to what motivated ProphetFuzz (CCS 2024), but here applied inside the directed-fuzzing loop rather than as an upstream option-selection oracle.

Method (inferred). Joint search over options × inputs, with option-space guidance folded into the DGF distance / seed-prioritisation criterion so that unfavourable option combinations are quickly abandoned and favourable ones are amplified toward the target location. The concrete scheduling formula, target set, baseline comparison, and CVE-reproduction numbers require the ACM PDF.

Why it stays on the radar. The option-aware seed-scheduling framing generalises to any DGF with a configurable target and is transferable to the fuzzing pipelines summarised in the [2026-07-06 digest](#). Top venue, followed-researcher provenance, and a genuinely reusable idea — a re-read is queued once the full text is retrievable.

Cross-Paper Synthesis

Two of today’s four proceeds are testing / fuzzing papers where the central move is to widen the search space to include a dimension that prior work had treated as fixed — NPC vehicle behavior in DynNPC (1.1) and command-line options in “It Takes Two” (2.2). Both papers make the same structural argument: the reason a class of scenario-based or directed-fuzzing tools was capped at a low true-positive rate is not that its search algorithm was weak but that its state space was under-defined. DynNPC’s headline efficiency numbers (72.18% EIV-share increase, 90.87% time-to-first-EIV reduction) are a direct empirical demonstration of what that reframing buys you when the missing dimension is central to the ground-truth notion of a bug; “It Takes Two” is aiming at the same phenomenon in a different substrate.

The two Outstanding papers frame today’s most interesting tension. DynNPC (1.1) trains a defender-side tool — a scenario generator that finds Ego-induced violations before deployment — and treats agents in the environment (NPC vehicles) as controllable adversaries to be steered by yielding/adversarial/overtaking policies. The Zhuo–Ding–Guo–Meng position paper (1.2) argues the opposite direction: for cybersecurity, the community must build offensive agents, not just defensive ones, because the alternative is asymmetric surrender to attackers who face no such constraint. Read together they sketch a common template: *use adversarial agents in a controlled environment (a simulation for DynNPC, an audited cyber range for 1.2) to generate the failure distribution that defensive systems will actually face in the wild*. The template is the same; the domain-specific translations are that “NPC” and “penetration-test agent” are the same abstract role, and “LGSVL simulator” and “audited cyber range” are the same abstract containment mechanism.

The CUDA paper (2.1) is the odd one out today — it is an empirical characterisation study rather than a search-space widening — but it fits the digest’s longer arc: it feeds the corpus of reproduced performance problems that downstream GPU-optimisation agents (of the kind surveyed in the

[2026-07-06 digest](#)) will need to be evaluated against, so it belongs in the same ecosystem even if it does not itself propose a new search or agent.

Writing & Rationale Insights

Two of today's four papers were only accessible at abstract depth because the ACM PDF endpoint blocks direct fetch and no arXiv mirror had been posted by the authors. The correct handling here is transparency: both papers get an explicit "access status" line in the private report and a "why it stays on the radar" framing in the public report, rather than a fabricated methods paragraph that would masquerade as first-hand reading. This keeps the calibration of the digest honest — when a card is labelled Outstanding, the reader can trust that the numbers and closing citable line came from the paper's own text; when it is labelled Keep with only positioning and expected shape, the reader knows the analysis is one abstraction level up. If ACM-gated papers become a persistent pattern (today marks two in one cycle), the pipeline should grow a third-fallback step that tries the corresponding author's personal page — Bihuan Chen's chenbihuan.github.io lists a full publication PDF archive — before demoting a paper to abstract-only. That is queued as a script change rather than a same-day action; the correct behaviour today is to publish the honest report and note the trigger.

The other lesson is about name-collision skips at Stage-1. The Arachne text-to-video paper surfaced through the Bihuan Chen alert but is almost certainly a different "B Chen" — the Fudan SE researcher's publication line does not contain text-to-video model training work. The private report calls this out as name-collision rather than off-topic to preserve the distinction. If a followed researcher genuinely pivots into an adjacent area, the digest should notice and Proceed; if the alert is picking up a homonym, the digest should skip. Today's card handling assumes the latter, but the private note explicitly flags the assumption so a later reader can second-guess it without having to re-derive the reasoning.